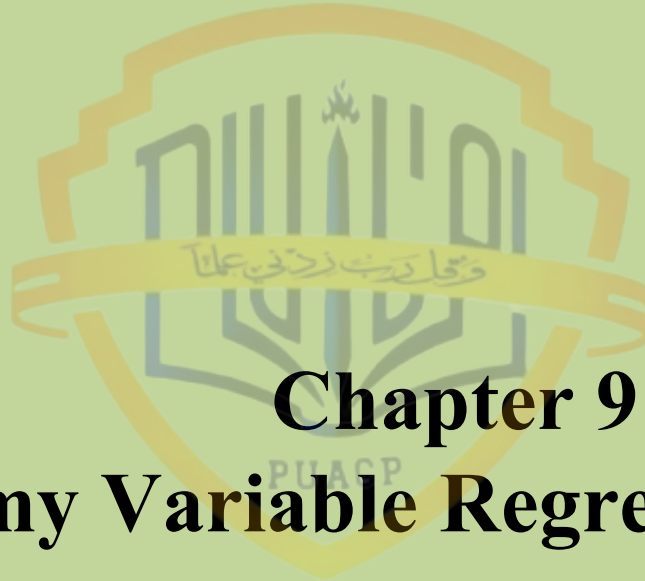




Chapter 9

Dummy Variable Regression Models

Damodar N. Gujarati and Dawn C Porter (2009)

THE NATURE OF DUMMY VARIABLES

- In this chapter, we consider models that may involve nominal scale variables. Such variables are also known as indicator variables, categorical variables, qualitative variables, or dummy variables.
- In regression analysis the dependent variable, or regressand, is frequently influenced not only by ratio scale variables (e.g., income, output, prices, costs, height, temperature) but also by variables that are essentially qualitative, or nominal scale, in nature, such as sex, race, color, religion, nationality, geographical region, political upheavals, and party affiliation.

The Nature of Dummy Variables

For Example

- Holding all other factors constant, female workers are found to earn less than their male counterparts.
- Nonwhite workers are found to earn less than whites.
- In these examples, the qualitative variables of gender and race seem to influence the regressand and clearly should be included among the explanatory variables, or the regressors.
- Such variables usually indicate **the presence or absence of a “quality” or an attribute**, such as male or female, black or white, Catholic or non-Catholic, Democrat or Republican, they are essentially nominal scale variables.
- We could “quantify” such attributes by constructing artificial variables that take on values of 1 or 0, 1 indicating the presence (or possession) of that attribute and 0 indicating the absence of that attribute.

- Variables that assume such 0 and 1 values are called *dummy variables*.
- Dummy variables can be incorporated in regression models just as easily as quantitative variables. As a matter of fact, a regression model may contain regressors that are all exclusively dummy, or qualitative, in nature are called *Analysis of Variance (ANOVA) models*.

9.2 ANOVA MODELS

Example 9.1 Public School Teachers' Salaries by Geographical Region

Table 9.1 gives data on average salary (in dollars) of public school teachers in 50 states and the District of Columbia for the academic year 2005–2006. These 51 areas are classified into three geographical regions:

- Northeast and North Central (21 states in all),
- South (17 states in all), and
- West (13 states in all).

Suppose we want to find out *if the average annual salary of public school teachers differs among the three geographical regions of the country*. If you take the simple arithmetic average of the average salaries of the teachers in the three regions, you will find that these averages for the three regions are as follows: \$49,538.71 (Northeast and North Central), \$46,293.59 (South), and \$48,104.62 (West).

9.2 ANOVA MODELS

- To see this, consider the following model:

$$Y_i = \beta_1 + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i \quad (9.2.1)$$

where Y_i = (average) salary of public school teacher in state i

D_{2i} = 1 if the state is in the Northeast or North Central
= 0 otherwise (i.e., in other regions of the country)

D_{3i} = 1 if the state is in the South
= 0 otherwise (i.e., in other regions of the country)

- The model (9.2.1) is a multiple regression model considered previously, except that, instead of quantitative regressors, we have only qualitative, or dummy, regressors taking the value of 1 if the observation belongs to a particular category and 0 if it does not belong to that category or group. Hereafter, we shall designate all dummy variables by the letter D . Table 9.1 shows the dummy variables thus constructed.

9.2 ANOVA MODELS

- Taking expectation of Eq. (9.2.1) on both sides, we obtain:
Mean salary of public school teachers in the Northeast and North Central.

$$E(Y_i | D_{2i} = 1, D_{3i} = 0) = \beta_1 + \beta_2 \quad (9.2.2)$$

Mean salary of public school teachers in the South:

$$E(Y_i | D_{2i} = 0, D_{3i} = 1) = \beta_1 + \beta_3 \quad (9.2.3)$$

Mean salary of public school teachers in the West:

$$E(Y_i | D_{2i} = 0, D_{3i} = 0) = \beta_1 \quad (9.2.4)$$

- The “slope” coefficients β_2 and β_3 tell by how much the mean salaries of teachers in the Northeast and North Central and in the South differ from the mean salary of teachers in the West. But how do we know if these differences are statistically significant? To answer this question, let us present the results based on the regression (9.2.1) using data given in Table 9.1:

9.2 ANOVA MODELS

TABLE 9.1 Average Salary of Public School Teachers by State, 2005–2006

	Salary	Spending	D_2	D_3		Salary	Spending	D_2	D_3
Connecticut	60,822	12,436	1	0	Georgia	49,905	8,534	0	1
Illinois	58,246	9,275	1	0	Kentucky	43,646	8,300	0	1
Indiana	47,831	8,935	1	0	Louisiana	42,816	8,519	0	1
Iowa	43,130	7,807	1	0	Maryland	56,927	9,771	0	1
Kansas	43,334	8,373	1	0	Mississippi	40,182	7,215	0	1
Maine	41,596	11,285	1	0	North Carolina	46,410	7,675	0	1
Massachusetts	58,624	12,596	1	0	Oklahoma	42,379	6,944	0	1
Michigan	54,895	9,880	1	0	South Carolina	44,133	8,377	0	1
Minnesota	49,634	9,675	1	0	Tennessee	43,816	6,979	0	1
Missouri	41,839	7,840	1	0	Texas	44,897	7,547	0	1
Nebraska	42,044	7,900	1	0	Virginia	44,727	9,275	0	1
New Hampshire	46,527	10,206	1	0	West Virginia	40,531	9,886	0	1
New Jersey	59,920	13,781	1	0	Alaska	54,658	10,171	0	0
New York	58,537	13,551	1	0	Arizona	45,941	5,585	0	0
North Dakota	38,822	7,807	1	0	California	63,640	8,486	0	0
Ohio	51,937	10,034	1	0	Colorado	45,833	8,861	0	0
Pennsylvania	54,970	10,711	1	0	Hawaii	51,922	9,879	0	0
Rhode Island	55,956	11,089	1	0	Idaho	42,798	7,042	0	0
South Dakota	35,378	7,911	1	0	Montana	41,225	8,361	0	0
Vermont	48,370	12,475	1	0	Nevada	45,342	6,755	0	0
Wisconsin	47,901	9,965	1	0	New Mexico	42,780	8,622	0	0
Alabama	43,389	7,706	0	1	Oregon	50,911	8,649	0	0
Arkansas	44,245	8,402	0	1	Utah	40,566	5,347	0	0
Delaware	54,680	12,036	0	1	Washington, D.C.	47,882	7,958	0	0
District of Columbia	59,000	15,508	0	1	Wyoming	50,692	11,596	0	0
Florida	45,308	7,762	0	1					

Note: $D_2 = 1$ for states in the Northeast and North Central; 0 otherwise.

$D_3 = 1$ for states in the South; 0 otherwise.

Source: National Educational Association, as reported in 2007.

9.2 ANOVA MODELS

- Taking expectation of Eq. (9.2.1) on both sides, we obtain: Mean salary of public school teachers in the Northeast and North Central.

$$E(Y_i | D_{2i} = 1, D_{3i} = 0) = \beta_1 + \beta_2 \quad (9.2.2)$$

Mean salary of public school teachers in the South:

$$E(Y_i | D_{2i} = 0, D_{3i} = 1) = \beta_1 + \beta_3 \quad (9.2.3)$$

Mean salary of public school teachers in the West:

$$E(Y_i | D_{2i} = 0, D_{3i} = 0) = \beta_1 \quad (9.2.4)$$

- The “slope” coefficients β_2 and β_3 tell by how much the mean salaries of teachers in the Northeast and North Central and in the South differ from the mean salary of teachers in the West. But how do we know if these differences are statistically significant? To answer this question, let us present the results based on the regression (9.2.1) using data given in Table 9.1:

$$\begin{array}{llll} \hat{Y}_i = 48,014.615 & + & 1,524.099D_{2i} & - & 1,721.027D_{3i} \\ \text{se} = (1857.204) & & (2363.139) & & (2467.151) \\ t = (25.853) & & (0.645) & & (-0.698) \\ & & (0.0000)^* & & (0.4888)^* & R^2 = 0.0440 \end{array} \quad (9.2.5)$$

9.2 ANOVA MODELS

- As these regression results show, the mean salary of teachers in the West is about \$48,015, that of teachers in the Northeast and North Central is higher by about \$1,524, and that of teachers in the South is lower by about \$1,721.
- The actual mean salaries in the last two regions can be easily obtained by adding these differential salaries to the mean salary of teachers in the West, as shown in Eqs. (9.2.3) and (9.2.4). Doing this, we will find that the mean salaries in the latter two regions are about \$49,539 ($48014.615 + 1524.099$) and \$46,294 ($48014.615 - 1721.027$).
- To see either mean salaries are statistically different from the mean salary of teachers in the West, the comparison category? We find out if each of the “slope” coefficients in Eq. (9.2.5) is statistically significant.

9.2 ANOVA MODELS

- The estimated slope coefficient for Northeast and North Central is not statistically significant, as their p value are 52 percent and 49 percent.
- Therefore, the overall conclusion is that statistically the mean salaries of public school teachers in the West, the Northeast and North Central, and the South are about the same. Diagrammatically, the situation is shown in Figure 9.1.

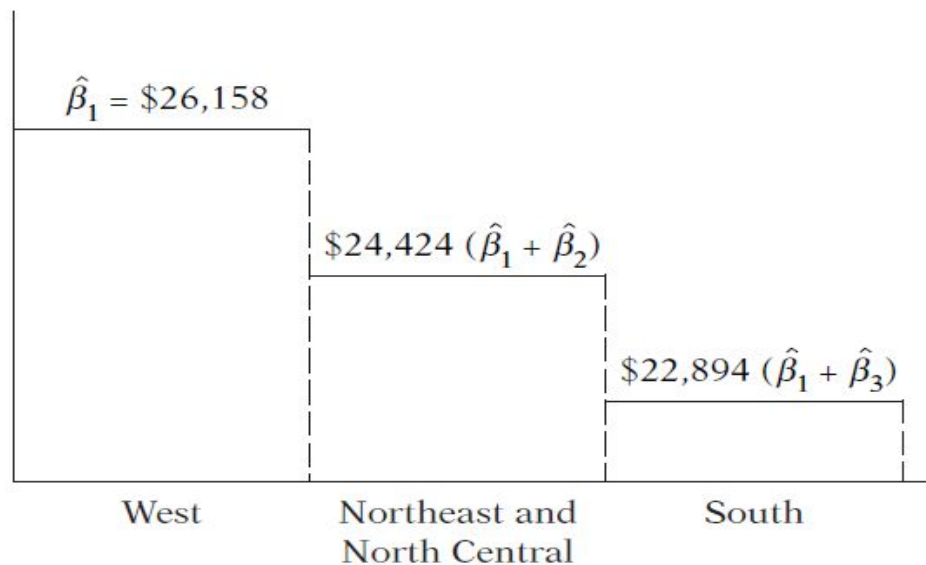


FIGURE 9.1

Average salary (in dollars) of public school teachers in three regions.

9.2 ANOVA MODELS

- The dummy variables will simply point out the differences, if they exist, but they do not suggest the reasons for the differences.
- Differences in educational levels, cost of living indexes, gender, and race may all have some effect on the observed differences. Therefore, unless we take into account all the other variables that may affect a teacher's salary, we will not be able to identify the cause(s) of the differences.

Caution in the Use of Dummy Variables

- Although they are easy to incorporate in the regression models, one must use the dummy variables carefully. In particular, consider the following aspects:

1. In Example 9.1, to distinguish the three regions, we used only two dummy variables, D2 and D3. Why did we not use three dummies to distinguish the three regions? Suppose we do that and write the model (9.2.1) as:

$$Y_i = \alpha + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i \quad (9.2.6)$$

- We now have a dummy variable for each of the three geographical regions.
- If you were to run the regression (9.2.6), the computer would “*refuse*” to run the regression (try it). Why?

9.2 ANOVA MODELS

- The reason is that in the setup of Eq. (9.2.6) where you have a dummy variable for each category or group and also an intercept, you have a case of perfect collinearity, that is, exact linear relationships among the variables.
- Refer to Table 9.1. Imagine that now we add the D1 column, taking the value of 1 whenever a state is in the West and 0 otherwise.
- Now if you add the three D columns horizontally, you will obtain a column that has 51 ones in it.
- In other words, the sum of the three D columns will simply reproduce the intercept column, thus leading to perfect collinearity. In this case, estimation of the model (9.2.6) is impossible.

9.2 ANOVA MODELS

1. Thus if a qualitative variable has m categories, introduce only $(m - 1)$ dummy variables. In our example, since the qualitative variable “region” has three categories, we introduced only two dummies. If we do not follow this rule, we will fall into what is called the **dummy variable trap (DVT)**. It is the situation of perfect collinearity or perfect multicollinearity, if there is more than one exact relationship among the variables.
2. The category for which no dummy variable is assigned is known as the *base*, *benchmark*, *control*, *comparison*, *reference*, or *omitted category*. And all comparisons are made in relation to the benchmark category.
3. The intercept value (β_1) represents the mean value of the benchmark category (west in Example 9.1). Hence, in the regression (9.2.5) the intercept value of about 48,015 represents the mean salary of teachers in the Western states.

9.2 ANOVA MODELS

4. The coefficients attached to the dummy variables in Eq. (9.2.1) are known as the *differential intercept coefficients* because they tell by how much the value of the category that receives the value of 1 differs from the intercept coefficient of the benchmark category. For example, in Eq. (9.2.5), the value of about 1,524 tells us that the mean salary of teachers in the Northeast or North Central is larger by about \$1,524 than the mean salary of about \$48,015 for the benchmark category, the West.

5. The choice of the benchmark category is **strictly up to the researcher**. Sometimes the choice of the benchmark is dictated by the particular problem at hand. Changing the benchmark category don't change the overall conclusion of our example (**why?**).

Consider south as base category, in this case, the intercept value will be about \$46,294, which is the mean salary of teachers in the South.

9.2 ANOVA MODELS

6. There is a way to avoid the dummy trap by introducing as many dummy variables as the number of categories of that variable, provided we do not introduce the intercept in such a model. Thus, if we drop the intercept term from Eq. (9.2.6), and consider the following model:

$$Y_i = \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i \quad (9.2.7)$$

- We do not fall into the dummy variable trap, as there is no longer perfect collinearity. But make sure that when you run this regression, you use the **no-intercept option** in your regression package.
- How do we interpret regression (9.2.7)? If you take the expectation of Eq. (9.2.7), you will find that:
- β_1 , β_2 , and β_3 are mean salaries for teachers in the West, Northeast and North Central, and south respectively.

9.2 ANOVA MODELS

The results of Eq. (9.2.7) for our illustrative example are as follows:

$$\begin{array}{llll} \hat{Y}_i = & 48,014.62D_{1i} & + 49,538.71D_{2i} & + 46,293.59D_{3i} \\ \text{se} = & (1857.204) & (1461.240) & (1624.077) \\ t = & (25.853)^* & (33.902)^* & (28.505)^* \\ & & & R^2 = 0.044 \end{array} \quad (9.2.8)$$

where * indicates that the p values of these t ratios are very small. As you can see, the dummy coefficients give directly the mean (salary) values in the three regions.

Which is a better method of introducing a dummy variable:

- (1) introduce a dummy for each category and omit the intercept term
- (2) Include the intercept term and introduce only $(m - 1)$ dummies, where m is the number of categories of the dummy variable?

9.2 ANOVA MODELS

As **Kennedy** notes: Most researchers find the equation with an intercept more convenient because

- It allows them to address more easily the questions in which they usually have the most interest, namely, whether or not the categorization makes a difference, and if so, by how much.
- If the categorization does make a difference, by how much is measured directly by the dummy variable coefficient estimates.
- Testing whether or not the categorization is relevant can be done by running a t test of a dummy variable coefficient against zero (or, to be more general, an F test on the appropriate set of dummy variable coefficient estimates).
- *Reference: Peter Kennedy, A Guide to Econometrics, 4th ed., MIT Press, Cambridge, Mass., 1998, p. 223.*

9.3 ANOVA MODELS

WITH TWO QUALITATIVE VARIABLES

Now we consider another ANOVA model, but with two qualitative variables.

EXAMPLE 9.2

*Hourly Wages in
Relation to
Marital Status
and Region of
Residence*

From a sample of 528 persons in May 1985, the following regression results were obtained.⁸

$$\begin{aligned}\hat{Y}_i &= 8.8148 + 1.0997D_{2i} - 1.6729D_{3i} \\ \text{se} &= (0.4015) \quad (0.4642) \quad (0.4854) \\ t &= (21.9528) \quad (2.3688) \quad (-3.4462) \\ &\quad (0.0000)^* \quad (0.0182)^* \quad (0.0006)^* \\ R^2 &= 0.0322\end{aligned}$$

(9.3.1)

where Y = hourly wage (\$)

D_2 = marital status; 1 = married, 0 = otherwise

D_3 = region of residence; 1 = South, 0 = otherwise

and * denotes the p values.

- In this example we have two qualitative regressors, each with two categories. Hence we have assigned a single dummy variable for each category.

9.3 ANOVA MODELS WITH TWO QUALITATIVE VARIABLES

- Which is the benchmark category here? Obviously, it is unmarried, non-South residence.
- The mean hourly wage in this benchmark is about \$8.81.
- Compared with this, the average hourly wage of those who are married is higher by about \$1.10, for an actual average wage of \$9.91 ($= 8.81 + 1.10$). By contrast, for those who live in the South, the average hourly wage is lower by about \$1.67, for an actual average hourly wage of \$7.14.
- All the differential intercepts are statistically significant, as their p values are quite low.
- *Once you go beyond one qualitative variable, you have to pay close attention to the category that is treated as the base category, since all comparisons are made in relation to that category. This is especially important when you have several qualitative regressors, each with several categories.*

9.4 Regression With A Mixture Of Quantitative And Qualitative Regressors: THE ANCOVA MODELS

- Typically, in most economic research a regression model contain some explanatory variables that are quantitative and some that are qualitative.
- Regression models containing an admixture of quantitative and qualitative variables are called **analysis of covariance (ANCOVA)** models.
- ANCOVA models are an extension of the ANOVA models in that they provide a method of statistically controlling the effects of quantitative regressors, called covariates or control variables, in a model that includes both quantitative and qualitative/dummy, regressors.

9.4 THE ANCOVA Models (Example 9.3)

- Let us reconsider Example 9.1 by maintaining that the average salary of public school teachers may not be different in the three regions if we take into account any variables that cannot be standardized across the regions. Consider, for example, the variable *expenditure on public schools by local authorities*, as public education is primarily a local and state question. To see if this is the case, we develop the following model:

$$Y_i = \beta_1 + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_i + u_i \quad (9.4.1)$$

where Y_i = average annual salary of public school teachers in state (\$)

X_i = spending on public school per pupil (\$)

$D_{2i} = 1$, if the state is in the Northeast or North Central
= 0, otherwise

$D_{3i} = 1$, if the state is in the South
= 0, otherwise

9.4 THE ANCOVA Models (Example 9.3)

From the data in Table 9.1, the results of the model (9.4.1) are as follows:

$$\begin{array}{rcccc}
 \hat{Y}_i = & 28,694.918 & - 2,954.127D_{2i} & - 3,112.194D_{3i} & + 2.3404X_i \\
 \text{se} = & (3262.521) & (1862.576) & (1819.873) & (0.3592) \\
 t = & (8.795)^* & (-1.586)^{**} & (-1.710)^{**} & (6.515)^*
 \end{array}
 \tag{9.4.2}$$

$R^2 = 0.4977$

where * indicates p values less than 5 percent, and ** indicates p values greater than 5 percent.

- *Ceteris paribus*: as public expenditure goes up by a dollar, on average, a public school teacher's salary goes up by about \$2.34.
- Controlling for spending on education, we now see that the differential intercept coefficient is not significant for either the Northeast and North Central region or for the South. These results are different from those of Eq. (9.2.5) where we did not account for the covariate, differences in per pupil public spending on education.

As these results suggest, *ceteris paribus*:

- as public expenditure goes up by a dollar, on average, a public school teacher's salary goes up by about \$2.34. Controlling for spending on education, we now see that the differential intercept coefficient is not significant for either the Northeast and North Central region or for the South.
- These results are different from those of Eq. (9.2.5). But this should not be surprising, for in Eq. (9.2.5) we did not account for the covariate, differences in per pupil public spending on education. Diagrammatically, we have the situation shown in Figure 9.2.
- Note that although we have shown three regression lines for the three regions, statistically the regression lines are the same for all three regions. Also note that the three regression lines are drawn parallel. (Why?)

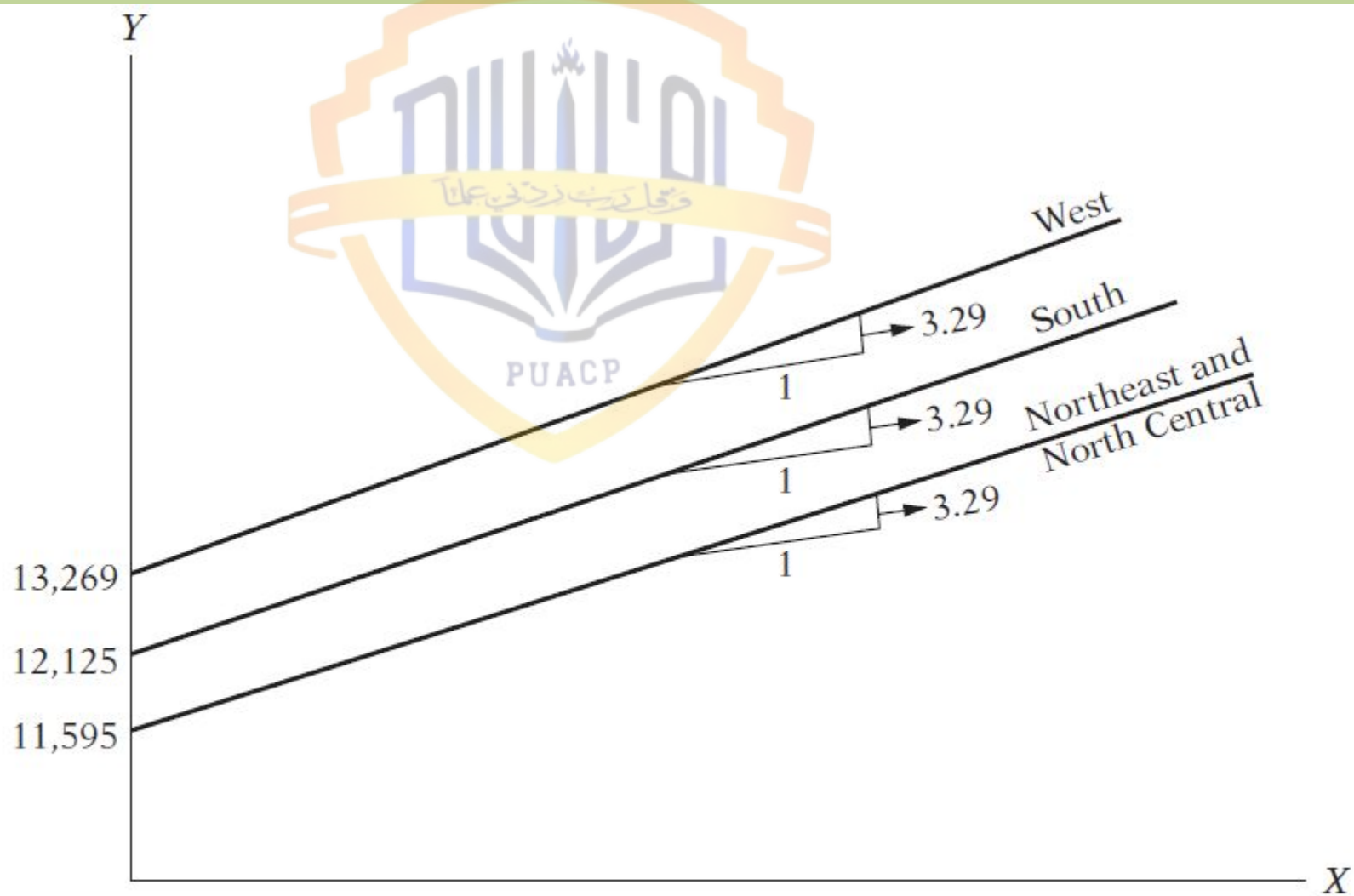


FIGURE 9.2

Public school teacher's salary (Y) in relation to per pupil expenditure on education (X).

9.5 The Dummy Variable Alternative to The Chow Test

- In Section 8.7 we discussed the Chow test to examine the structural stability of a regression model. The example we discussed there related to the relationship between savings and income in the United States over the period 1970–1995.
- We divided the sample period into two, 1970–1981 and 1982–1995, and showed on the basis of the Chow test that there was a difference in the regression of savings on income between the two periods.
- However, we could not tell whether the difference in the two regressions was because of differences in the intercept terms or the slope coefficients or both. Very often this knowledge itself is very useful.

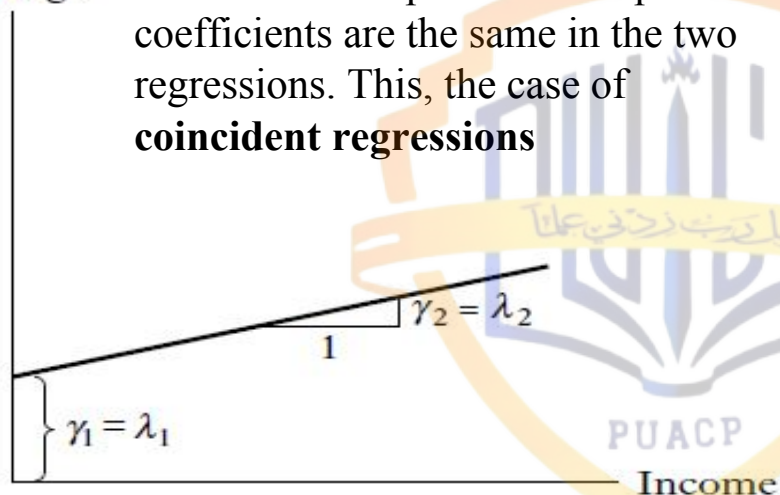
$$\text{Time period 1970–1981: } Y_t = \lambda_1 + \lambda_2 X_t + u_{1t} \quad n_1 = 12 \quad (8.8.1)$$

$$\text{Time period 1982–1995: } Y_t = \gamma_1 + \gamma_2 X_t + u_{2t} \quad n_2 = 14 \quad (8.8.2)$$

$$\text{Time period 1970–1995: } Y_t = \alpha_1 + \alpha_2 X_t + u_t \quad n = (n_1 + n_2) = 26 \quad (8.8.3)$$

Savings

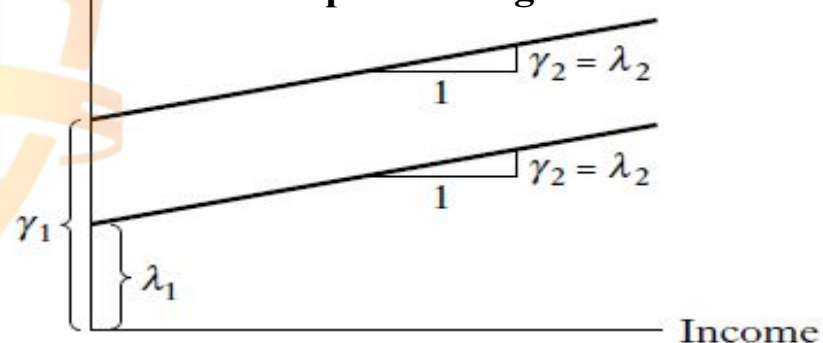
Both the intercept and the slope coefficients are the same in the two regressions. This, the case of **coincident regressions**



(a) Coincident regressions

Savings

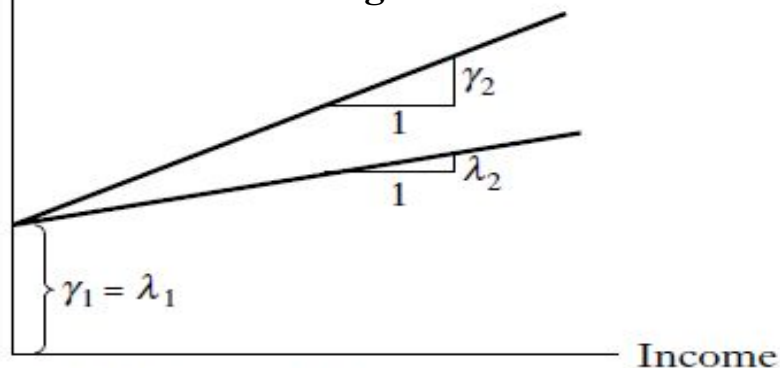
Only the intercepts in the two regressions are different but the slopes are the same. This is the case of **parallel regressions**



(b) Parallel regressions

Savings

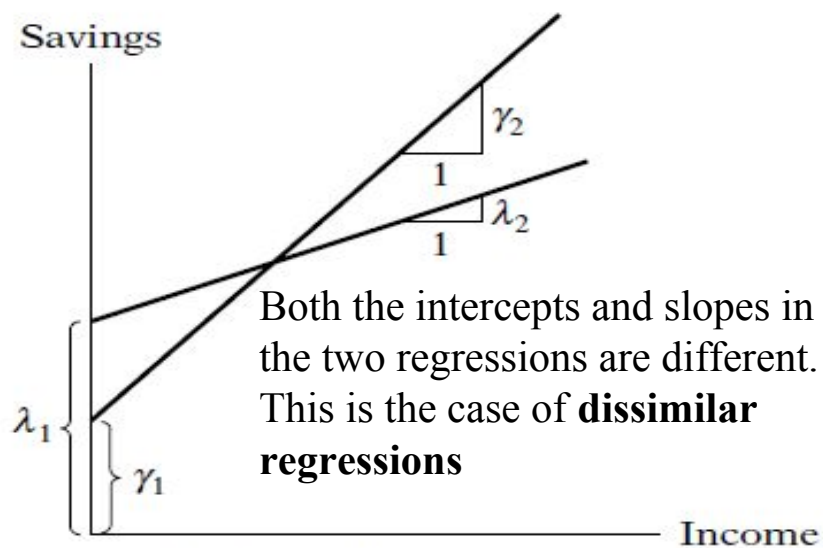
The intercepts in the two regressions are the same, but the slopes are different. This is the situation of **concurrent regressions**



(c) Concurrent regressions

Savings

Both the intercepts and slopes in the two regressions are different. This is the case of **dissimilar regressions**



(d) Dissimilar regressions

FIGURE 9.3 Plausible savings–income regressions.

9.5 THE DUMMY VARIABLE ALTERNATIVE TO THE CHOW TEST

Observation	Savings	Income	Dum
1970	61	727.1	0
1971	68.6	790.2	0
1972	63.6	855.3	0
1973	89.6	965	0
1974	97.6	1054.2	0
1975	104.4	1159.2	0
1976	96.4	1273	0
1977	92.5	1401.4	0
1978	112.6	1580.1	0
1979	130.1	1769.5	0
1980	161.8	1973.3	0
1981	199.1	2200.2	0
1982	205.5	2347.3	1
1983	167	2522.4	1
1984	235.7	2810	1
1985	206.2	3002	1
1986	196.5	3187.6	1
1987	168.4	3363.1	1
1988	189.1	3640.8	1
1989	187.8	3894.5	1
1990	208.7	4166.8	1
1991	246.4	4343.7	1
1992	272.6	4613.7	1
1993	214.4	4790.2	1
1994	189.4	5021.7	1
1995	249.3	5320.8	1

Note: Dum = 1 for observations beginning in 1982; 0 otherwise.
Savings and income figures are in billions of dollars.

9.5 THE DUMMY VARIABLE ALTERNATIVE TO THE CHOW TEST

$$Y_t = \alpha_1 + \alpha_2 D_t + \beta_1 X_t + \beta_2 (D_t X_t) + u_t \quad (9.5.1)$$

To see the implications of Eq. (9.5.1), and, assuming, as usual, that $E(u_i) = 0$, we obtain:

Mean savings function for 1970–1981:

$$E(Y_t | D_t = 0, X_t) = \alpha_1 + \beta_1 X_t \quad (9.5.2)$$

Mean savings function for 1982–1995:

$$E(Y_t | D_t = 1, X_t) = (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) X_t \quad (9.5.3)$$

The reader will notice that these are the same functions as Eqs. (8.7.1) and (8.7.2), with $\lambda_1 = \alpha_1$, $\lambda_2 = \beta_1$, $\gamma_1 = (\alpha_1 + \alpha_2)$, and $\gamma_2 = (\beta_1 + \beta_2)$. Therefore, estimating Eq. (9.5.1) is equivalent to estimating the two individual savings functions in Eqs. (8.7.1) and (8.7.2).

9.5 THE DUMMY VARIABLE ALTERNATIVE TO THE CHOW TEST

$$Y_t = \alpha_1 + \alpha_2 D_t + \beta_1 X_t + \beta_2 (D_t X_t) + u_t \quad (9.5.1)$$

- In Eq. (9.5.1), α_2 is the **differential intercept**, as previously, and β_2 is the **differential slope coefficient** (also called the **slope drifter**), indicating by how much the slope coefficient of the second period's savings function differs from that of the first period.
- Notice how the introduction of the dummy variable D in the **interactive, or multiplicative, form** (D multiplied by X) enables us to differentiate between slope coefficients of the two periods, just as the introduction of the dummy variable in the **additive form** enabled us to distinguish between the intercepts of the two periods.

9.5 THE DUMMY VARIABLE ALTERNATIVE TO THE CHOW TEST

Before we proceed further, let us first present the regression results of model (9.5.1) applied to the U.S. savings–income data.

$$\begin{array}{rcccc} \hat{Y}_t = & 1.0161 & + 152.4786D_t + & 0.0803X_t - & 0.0655(D_tX_t) \\ \text{se} = & (20.1648) & (33.0824) & (0.0144) & (0.0159) \\ t = & (0.0504)^{**} & (4.6090)^* & (5.5413)^* & (-4.0963)^* \end{array} \quad (9.5.4)$$

$R^2 = 0.8819$

where * indicates p values less than 5 percent and ** indicates p values greater than 5 percent.

As these regression results show, both the differential intercept and slope coefficients are statistically significant, strongly suggesting that the savings–income regressions for the two time periods are different, as in Figure 9.3d.

Savings–income regression, 1970–1981:

$$\hat{Y}_t = 1.0161 + 0.0803X_t \quad (9.5.5)$$

Savings–income regression, 1982–1995:

$$\begin{aligned} \hat{Y}_t &= (1.0161 + 152.4786) + (0.0803 - 0.0655)X_t \\ &= 153.4947 + 0.0148X_t \end{aligned} \quad (9.5.6)$$

9.5 THE DUMMY VARIABLE ALTERNATIVE TO THE CHOW TEST

The advantages of the dummy variable technique over the Chow test (i.e., estimating the three regressions [8.7.1], [8.7.2], and [8.7.3]) can now be seen readily:

1. We need to run only a single regression because the individual regressions can easily be derived from it in the manner indicated by equations (9.5.2) and (9.5.3).
2. The single regression (9.5.1) can be used to test a variety of hypotheses.
3. The Chow test does not explicitly tell us *which* coefficient, intercept, or slope is different, or whether (as in this example) both are different in the two periods.

9.6 INTERACTION EFFECTS USING DUMMY VARIABLES

- Dummy variables are a flexible tool that can handle a variety of interesting problems. To see this, consider the following model:

$$Y_i = \alpha_1 + \alpha_2 D_{2i} + \alpha_3 D_{3i} + \beta X_i + u_i \quad (9.6.1)$$

where

Y = hourly wage in dollars

X = education (years of schooling)

$D_2 = 1$ if female, 0 otherwise

$D_3 = 1$ if nonwhite and non-Hispanic, 0 otherwise

- In this model gender and race are qualitative regressors and education is a quantitative regressor.
- Implicit in this model is the assumption that the differential effect of the gender dummy D_2 is constant across the two categories of race and the differential effect of the race dummy D_3 is also constant across the two sexes. That is to say, if the mean salary is higher for males than for females, this is so whether they are nonwhite/non-Hispanic or not.

9.6 INTERACTION EFFECTS USING DUMMY VARIABLES

- Likewise, nonwhite/non-Hispanics have lower mean wages, this is so whether they are females or males.
- In many applications such an assumption may be invalid. A female nonwhite/non-Hispanic may earn lower wages than a male nonwhite/non-Hispanic. In other words, there may be **interaction** between the two qualitative variables D_2 and D_3 . Therefore their effect on mean Y may not be simply **additive** as in Eq. (9.6.1) but **multiplicative** as well, as in the following model.

$$Y_i = \alpha_1 + \alpha_2 D_{2i} + \alpha_3 D_{3i} + \alpha_4 (D_{2i} D_{3i}) + \beta X_i + u_i \quad (9.6.2)$$

where the variables are as defined for model (9.6.1).

From Eq. (9.6.2), we obtain:

$$E(Y_i | D_{2i} = 1, D_{3i} = 1, X_i) = (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) + \beta X_i \quad (9.6.3)$$

which is the mean hourly wage function for female nonwhite/non-Hispanic workers. Observe that

α_2 = differential effect of being a female

α_3 = differential effect of being a nonwhite/non-Hispanic

α_4 = differential effect of being a female nonwhite/non-Hispanic

9.6 INTERACTION EFFECTS USING DUMMY VARIABLES

- It shows that the mean hourly wages of female nonwhite/non-Hispanics is different (by α_4) from the mean hourly wages of females or nonwhite/non-Hispanics.
- If, for instance, all the three differential dummy coefficients are negative, this would imply that female nonwhite/non-Hispanic workers earn much lower mean hourly wages than female or nonwhite/non-Hispanic workers as compared with the base category, which in the present example is male white or Hispanic.
- Now the reader can see how the **interaction dummy** (i.e., the product of two qualitative or dummy variables) modifies the effect of the two attributes considered individually (i.e. additively).

9.6 INTERACTION EFFECTS USING DUMMY VARIABLES

EXAMPLE 9.5

*Average Hourly
Earnings in
Relation to
Education,
Gender, and
Race*

Let us first present the regression results based on model (9.6.1). Using the data that were used to estimate regression (9.3.1), we obtained the following results:

$$\begin{aligned}\hat{Y}_i = & -0.2610 - 2.3606D_{2i} - 1.7327D_{3i} + 0.8028X_i \\ t = & (-0.2357)^{**} \quad (-5.4873)^* \quad (-2.1803)^* \quad (9.9094)^* \quad (9.6.4) \\ & R^2 = 0.2032 \quad n = 528\end{aligned}$$

where * indicates p values less than 5 percent and ** indicates p values greater than 5 percent.

(Continued)

We now consider the results of model (9.6.2), which includes the interaction dummy.

$$\begin{aligned}\hat{Y}_i = & -0.26100 - 2.3606D_{2i} - 1.7327D_{3i} + 2.1289D_{2i}D_{3i} + 0.8028X_i \\ t = & (-0.2357)^{**} \quad (-5.4873)^* \quad (-2.1803)^* \quad (1.7420)^{**} \quad (9.9095)^{**} \quad (9.6.5) \\ & R^2 = 0.2032 \quad n = 528\end{aligned}$$

where * indicates p values less than 5 percent and ** indicates p values greater than 5 percent.

9.6 INTERACTION EFFECTS USING DUMMY VARIABLES

- The two additive dummies are still statistically significant, but the interactive dummy is not at the conventional 5 percent level.
- Holding the level of education constant, if we add the three dummy coefficients we will obtain: -1.964 ($= -2.3605 - 1.7327 + 2.1289$), which means that mean hourly wages of nonwhite/non-Hispanic female workers is lower by about \$1.96, which is between the value of -2.3605 (gender difference alone) and -1.7327 (race difference alone).

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

- Many economic time series based on monthly or quarterly data exhibit seasonal patterns (regular oscillatory movements). Examples are sales of department stores at Christmas and other major holiday times, demand for money (or cash balances) by households at holiday times, demand for ice cream and soft drinks during summer, prices of crops right after harvesting season, demand for air travel, etc.
- Often it is desirable to remove the seasonal factor, or *component*, from a time series so that one can concentrate on the other components, such as the trend.
- The process of removing the seasonal component from a time series is known as **deseasonalization** or **seasonal adjustment**, and the time series thus obtained is called the **deseasonalized**, or **seasonally adjusted**, time series.
- Important economic time series, such as the unemployment rate, the consumer price index (CPI), the producer's price index (PPI), and the index of industrial production, are usually published in seasonally adjusted form.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

- There are several methods of deseasonalizing a time series, but we will consider only one of these methods, namely, the *method of dummy variables*.
- To illustrate how the dummy variables can be used to deseasonalize economic time series, consider the data given in Table 9.3. This table gives quarterly data for the years 1978–1995 on the sale of four major appliances, dishwashers, garbage disposers, refrigerators, and washing machines, all data in thousands of units. The table also gives data on durable goods expenditure in 1982 billions of dollars.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

DISH	DISP	FRIG	WASH	DUR	DISH	DISP	FRIG	WASH	DUR
841	798	1317	1271	252.6	480	706	943	1036	247.7
957	837	1615	1295	272.4	530	582	1175	1019	249.1
999	821	1662	1313	270.9	557	659	1269	1047	251.8
960	858	1295	1150	273.9	602	837	973	918	262
894	837	1271	1289	268.9	658	867	1102	1137	263.3
851	838	1555	1245	262.9	749	860	1344	1167	280
863	832	1639	1270	270.9	827	918	1641	1230	288.5
878	818	1238	1103	263.4	858	1017	1225	1081	300.5
792	868	1277	1273	260.6	808	1063	1429	1326	312.6
589	623	1258	1031	231.9	840	955	1699	1228	322.5
657	662	1417	1143	242.7	893	973	1749	1297	324.3
699	822	1185	1101	248.6	950	1096	1117	1198	333.1
675	871	1196	1181	258.7	838	1086	1242	1292	344.8
652	791	1410	1116	248.4	884	990	1684	1342	350.3
628	759	1417	1190	255.5	905	1028	1764	1323	369.1
529	734	919	1125	240.4	909	1003	1328	1274	356.4

Note: DISH = dishwashers; DISP = garbage disposers; FRIG = refrigerators; WASH = washing machines; DUR = durable goods expenditure, billions of 1982 dollars.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

- We consider only the sales of refrigerators over the sample period. To see if there is seasonal pattern we consider the following model:

$$Y_t = \alpha_1 D_{1t} + \alpha_2 D_{2t} + \alpha_3 D_{3t} + \alpha_4 D_{4t} + u_t \quad (9.7.1)$$

- where Y_t = sales of refrigerators (in thousands) and the D 's are the dummies, taking a value of 1 in the relevant quarter and 0 otherwise.
- *Note that to avoid the dummy variable trap, we are assigning a dummy to each quarter of the year, but omitting the intercept term.*
- If there is any seasonal effect in a given quarter, that will be indicated by a statistically significant t value of the dummy coefficient for that quarter.
- The coefficient of each quarter will give us the mean refrigerator sales in each quarter or season

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

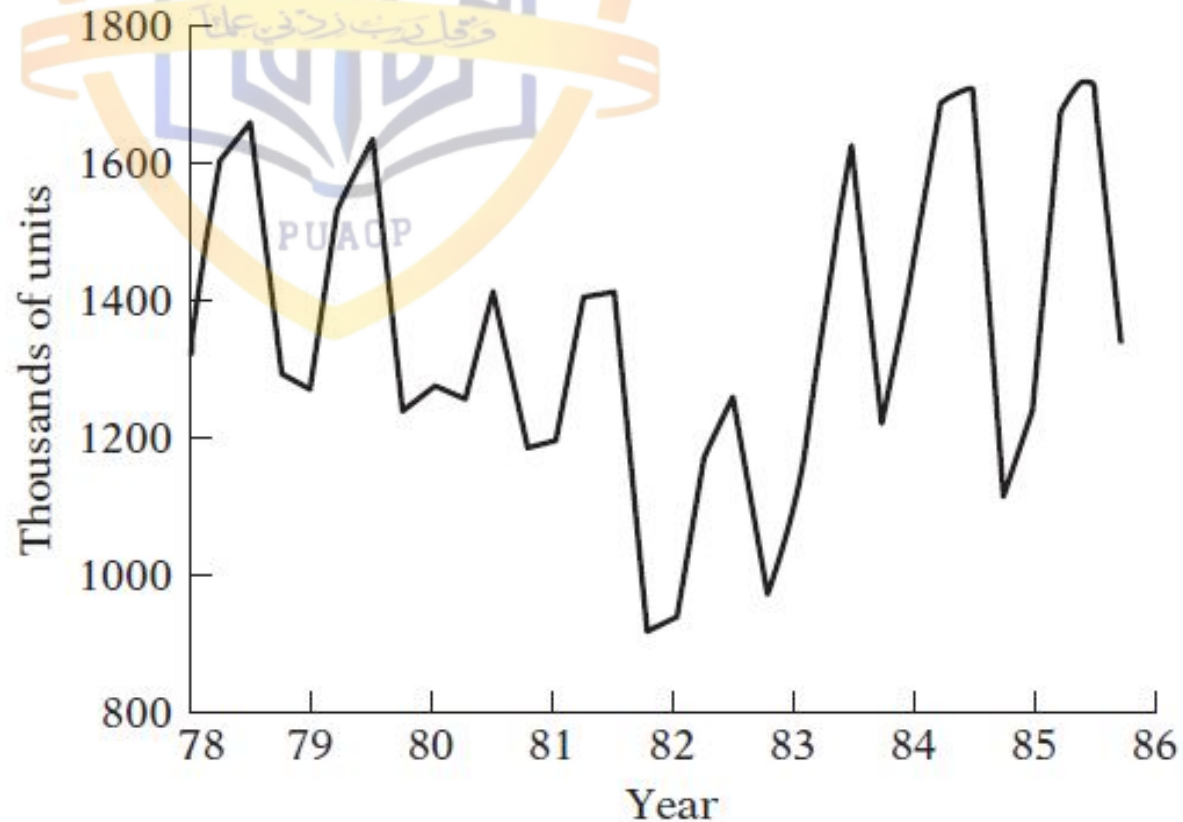


FIGURE 9.4 Sales of refrigerators 1978–1985 (quarterly).

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

From the data on refrigerator sales given in Table 9.4, we obtain the following regression results:

$$\begin{aligned} \hat{Y}_t &= 1,222.125D_{1t} + 1,467.500D_{2t} + 1,569.750D_{3t} + 1,160.000D_{4t} \\ t &= \quad (20.3720) \quad \quad (24.4622) \quad \quad (26.1666) \quad \quad (19.3364) \end{aligned} \quad (9.7.2)$$
$$R^2 = 0.5317$$

Note: We have not given the standard errors of the estimated coefficients, as each standard error is equal to 59.9904, because all the dummies take only a value of 1 or zero.

The estimated α coefficients in Eq. (9.7.2) represent the average, or *mean*, sales of refrigerators (in thousands of units) in each season (i.e., quarter). Thus, the average sale of refrigerators in the first quarter, in thousands of units, is about 1,222, that in the second quarter about 1,468, that in the third quarter about 1,570, and that in the fourth quarter about 1,160.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

- Now we treat the first quarter as the reference quarter and assign dummies to the second, third, and fourth quarters. This produces the following regression results:

$$\begin{array}{ccccccc} \hat{Y}_t = & 1,222.1250 & + & 245.3750D_{2t} & + & 347.6250D_{3t} & - & 62.1250D_{4t} \\ t = & (20.3720)^* & & (2.8922)^* & & (4.0974)^* & & (-0.7322)^{**} \\ & & & & & & & R^2 = 0.5318 \end{array} \quad (9.7.3)$$

where * indicates p values less than 5 percent and ** indicates p values greater than 5 percent.

- Eq. (9.7.3) shows that the average value of Y for the fourth quarter is not statistically different from the average value for the first quarter, as the dummy coefficient for the fourth quarter is not statistically significant.
- Of course, your answer will change, depending on which quarter you treat as the benchmark quarter, but the overall conclusion will not change.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

TABLE 9.4

**U.S. Refrigerator
Sales (thousands),
1978–1985
(quarterly)**

Source: *Business Statistics
and Survey of Current
Business*, Department of
Commerce (various issues).

FRIG	DUR	D_2	D_3	D_4	FRIG	DUR	D_2	D_3	D_4
1317	252.6	0	0	0	943	247.7	0	0	0
1615	272.4	1	0	0	1175	249.1	1	0	0
1662	270.9	0	1	0	1269	251.8	0	1	0
1295	273.9	0	0	1	973	262.0	0	0	1
1271	268.9	0	0	0	1102	263.3	0	0	0
1555	262.9	1	0	0	1344	280.0	1	0	0
1639	270.9	0	1	0	1641	288.5	0	1	0
1238	263.4	0	0	1	1225	300.5	0	0	1
1277	260.6	0	0	0	1429	312.6	0	0	0
1258	231.9	1	0	0	1699	322.5	1	0	0
1417	242.7	0	1	0	1749	324.3	0	1	0
1185	248.6	0	0	1	1117	333.1	0	0	1
1196	258.7	0	0	0	1242	344.8	0	0	0
1410	248.4	1	0	0	1684	350.3	1	0	0
1417	255.5	0	1	0	1764	369.1	0	1	0
919	240.4	0	0	1	1328	356.4	0	0	1

Note: FRIG = refrigerator sales, thousands.

DUR = durable goods expenditure, billions of 1982 dollars.

D_2 = 1 in the second quarter, 0 otherwise.

D_3 = 1 in the third quarter, 0 otherwise.

D_4 = 1 in the fourth quarter, 0 otherwise.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

- To obtain the deseasonalized time series of refrigerator sales we estimate the values of Y from model (9.7.2) (or [9.7.3]) for each observation and subtract them from the actual values of Y , that is, we obtain $(Y_t - \hat{Y}_t)$ which are simply the residuals from the regression (9.7.2) as shown in Table 9.5.15.
- To these residuals, we have to add the mean of the Y series to get the forecasted values.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

EXAMPLE 9.6
(Continued)

TABLE 9.5 Refrigerator Sales Regression: Actual, Fitted, and Residual Values (Eq. 9.7.3)

	Actual	Fitted	Residuals	Residuals Graph
				0
1978-I	1317	1222.12	94.875	. *
1978-II	1615	1467.50	147.500	. *
1978-III	1662	1569.75	92.250	. *
1978-IV	1295	1160.00	135.000	. *
1979-I	1271	1222.12	48.875	. *
1979-II	1555	1467.50	87.500	. *
1979-III	1639	1569.75	69.250	. *
1979-IV	1238	1160.00	78.000	. *
1980-I	1277	1222.12	54.875	. *
1980-II	1258	1467.50	-209.500	*.
1980-III	1417	1569.75	-152.750	*.
1980-IV	1185	1160.00	25.000	. *
1981-I	1196	1222.12	-26.125	. *
1981-II	1410	1467.50	-57.500	. *
1981-III	1417	1569.75	-152.750	. *
1981-IV	919	1160.00	-241.000	*.
1982-I	943	1222.12	-279.125	*.
1982-II	1175	1467.50	-292.500	*.
1982-III	1269	1569.75	-300.750	*.
1982-IV	973	1160.00	-187.000	*.
1983-I	1102	1222.12	-120.125	. *
1983-II	1344	1467.50	-123.500	. *
1983-III	1641	1569.75	71.250	. *
1983-IV	1225	1160.00	65.000	. *
1984-I	1429	1222.12	206.875	. *
1984-II	1699	1467.50	231.500	. *
1984-III	1749	1569.75	179.250	. *
1984-IV	1117	1160.00	-43.000	. *
1985-I	1242	1222.12	19.875	. *
1985-II	1684	1467.50	216.500	. *
1985-III	1764	1569.75	194.250	. *
1985-IV	1328	1160.00	168.000	. *
				- 0 +

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

Since models (9.7.2) and (9.7.3) do not contain any covariates, will the picture change if we bring in a quantitative regressor in the model? Since expenditure on durable goods has an important factor influence on the demand for refrigerators, let us expand our model (9.7.3) by bringing in this variable. The data for durable goods expenditure in billions of 1982 dollars are already given in Table 9.3. This is our (quantitative) X variable in the model. The regression results are as follows

$$\begin{aligned} \hat{Y}_t = & 456.2440 + 242.4976D_{2t} + 325.2643D_{3t} - 86.0804D_{4t} + 2.7734X_t \\ t = & (2.5593)^* \quad (3.6951)^* \quad (4.9421)^* \quad (-1.3073)^{**} \quad (4.4496)^* \quad \textbf{(9.7.4)} \\ & R^2 = 0.7298 \end{aligned}$$

where * indicates p values less than 5 percent and ** indicates p values greater than 5 percent.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

- Again, keep in mind that we are treating the first quarter as our base. As in Eq. (9.7.3), we see that the differential intercept coefficients for the second and third quarters are statistically different from that of the first quarter, but the intercepts of the fourth quarter and the first quarter are statistically about the same.
- The coefficient of X (durable goods expenditure) of about 2.77 tells us that, allowing for seasonal effects, if expenditure on durable goods goes up by a dollar, on average, sales of refrigerators go up by about 2.77 units, that is, approximately 3 units; bear in mind that refrigerators are in thousands of units and X is in (1982) billions of dollars.

9.7 THE USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

- An interesting question here is: Just as sales of refrigerators exhibit seasonal patterns, would not expenditure on durable goods also exhibit seasonal patterns?
- How then do we take into account seasonality in X ?
- The interesting thing about Eq. (9.7.4) is that the dummy variables in that model not only remove the seasonality in Y but also the seasonality, if any, in X . (This follows from a well-known theorem in statistics, known as the **Frisch–Waugh theorem**.) So to speak, we kill (deseasonalize) two birds (two series) with one stone (the dummy technique).
- If you want an informal proof of the preceding statement, just follow these steps: (1) Run the regression of Y on the dummies as in Eq. (9.7.2) or Eq. (9.7.3) and save the residuals, say, $S1$; these residuals represent deseasonalized Y . (2) Run a similar regression for X and obtain the residuals from this regression, say, $S2$; these residuals represent deseasonalized X . (3) Regress $S1$ on $S2$. You will find that the slope coefficient in this regression is precisely the coefficient of X in the regression (9.7.4).

9.8 PIECEWISE LINEAR REGRESSION

- To illustrate yet another use of dummy variables, consider Figure 9.5, which shows how a hypothetical company remunerates its sales representatives.
- It pays commissions based on sales in such a manner that up to a certain level, the target, or threshold, level X^* , there is one (stochastic) commission structure and beyond that level another. (Note: Besides sales, other factors affect sales commission.
- Assume that these other factors are represented by the stochastic disturbance term.) More specifically, it is assumed that sales commission increases linearly with sales until the threshold level X^* , after which it continues to increase linearly with sales but at a much steeper rate.
- Thus, we have a piecewise linear regression consisting of two linear pieces or segments, which are labeled I and II in Figure 9.5, and the commission function changes its slope at the threshold value.

9.8 PIECEWISE LINEAR REGRESSION

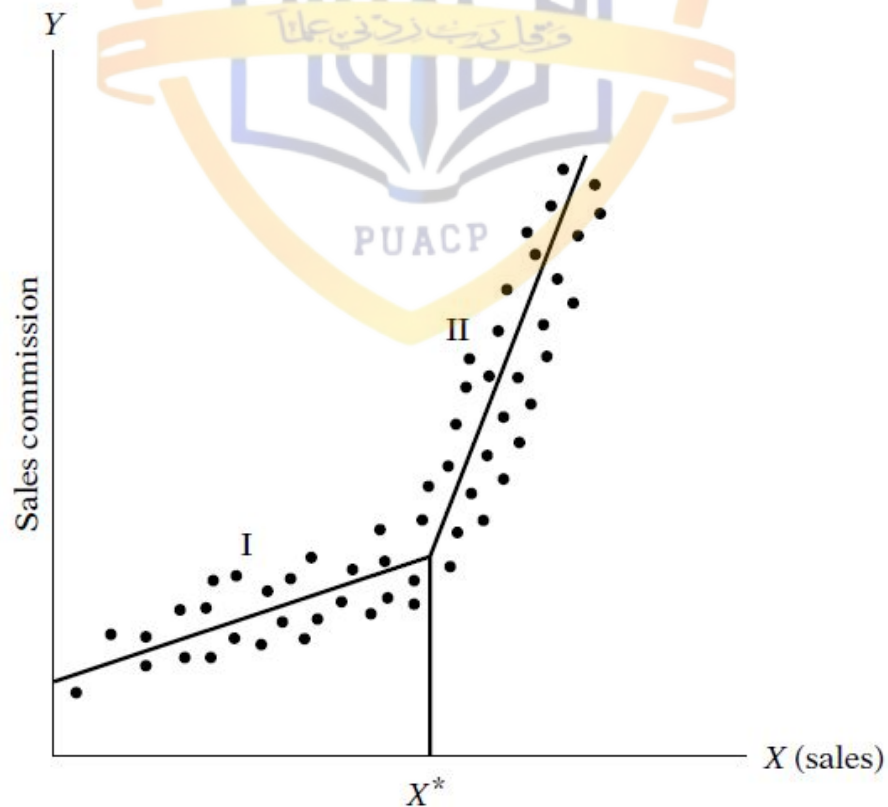


FIGURE 9.5 Hypothetical relationship between sales commission and sales volume. (Note: The intercept on the Y axis denotes minimum guaranteed commission.)

9.8 PIECEWISE LINEAR REGRESSION

- Given the data on commission, sales, and the value of the threshold level X^* , the technique of dummy variables can be used to estimate the (differing) slopes of the two segments of the piecewise linear regression shown in Figure 9.5.
- We proceed as follows:

$$Y_i = \alpha_1 + \beta_1 X_i + \beta_2 (X_i - X^*) D_i + u_i \quad (9.8.1)$$

where Y_i = sales commission

X_i = volume of sales generated by the sales person

X^* = threshold value of sales also known as a **knot** (known in advance)

$D = 1$ if $X_i > X^*$
 $= 0$ if $X_i < X^*$

9.8 PIECEWISE LINEAR REGRESSION

Assuming $E(u_i) = 0$, we see at once that

$$E(Y_i | D_i = 0, X_i, X^*) = \alpha_1 + \beta_1 X_i \quad (9.8.2)$$

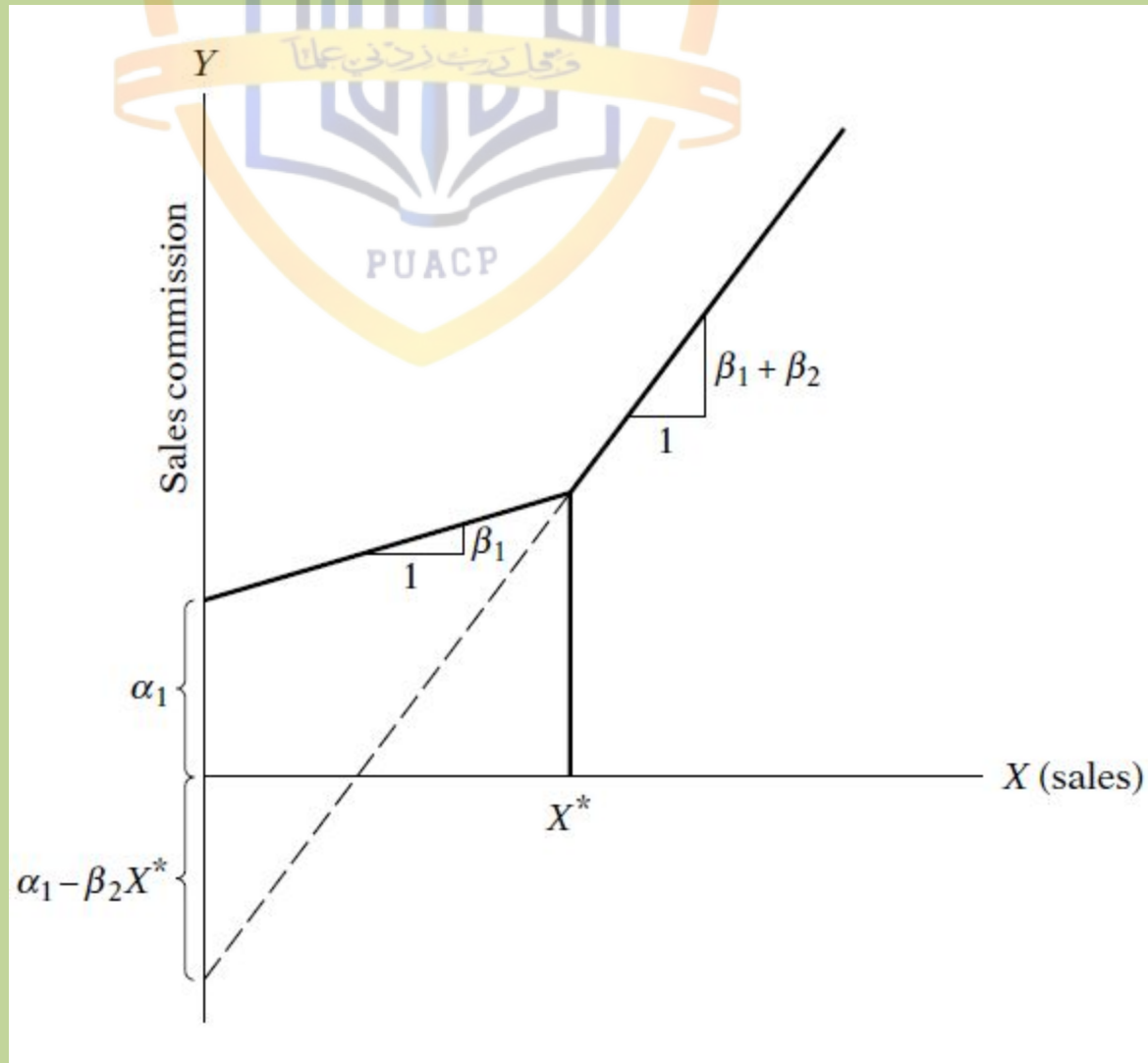
which gives the mean sales commission up to the target level X^* and

$$E(Y_i | D_i = 1, X_i, X^*) = \alpha_1 - \beta_2 X^* + (\beta_1 + \beta_2) X_i \quad (9.8.3)$$

which gives the mean sales commission beyond the target level X^* .

- Thus, β_1 gives the slope of the regression line in segment I, and $\beta_1 + \beta_2$ gives the slope of the regression line in segment II of the piecewise linear regression shown in Figure 9.5.
- A test of the hypothesis that there is no break in the regression at the threshold value X^* can be conducted easily by noting the statistical significance of the estimated differential slope coefficient $\widehat{\beta}_2$.
- Incidentally, the piecewise linear regression we have just discussed is an example of a more general class of functions known as **spline functions**.

FIGURE 9.6 Parameters of the piecewise linear regression.



EXAMPLE 9.7 TOTAL COST IN RELATION TO OUTPUT

- As an example of the application of the piecewise linear regression, consider the hypothetical total cost–total output data given in Table 9.6. We are told that the total cost may change its slope at the output level of 5,500 units.

Letting Y in Eq. (9.8.4) represent total cost and X total output, we obtain the following results:

$$\begin{aligned} \hat{Y}_i &= -145.72 + 0.2791X_i + 0.0945(X_i - X_i^*)D_i \\ t &= (-0.8245) \quad (6.0669) \quad (1.1447) \\ R^2 &= 0.9737 \quad X^* = 5,500 \end{aligned} \quad (9.8.4)$$

As these results show, the marginal cost of production is about 28 cents per unit and although it is about 37 cents (28 + 9) for output over 5,500 units, the difference between the two is not statistically significant because the dummy variable is not significant at, say, the 5 percent level. For all practical purposes, then, one can regress total cost on total output, dropping the dummy variable.

Total Cost, Dollars	Output, Units
256	1,000
414	2,000
634	3,000
778	4,000
1,003	5,000
1,839	6,000
2,081	7,000
2,423	8,000
2,734	9,000
2,914	10,000

9.10 SOME TECHNICAL ASPECTS OF THE DUMMY VARIABLE TECHNIQUE

The Interpretation of Dummy Variables in Semilogarithmic Regressions

In Chapter 6 we discussed the log–lin models, where the regressand is logarithmic and the regressors are linear. In such a model, the slope coefficients of the regressors give the *semi*elasticity, that is, the percentage change in the regressand for a unit change in the regressor. *This is only so if the regressor is quantitative.* What happens if a regressor is a dummy variable? To be specific, consider the following model:

$$\ln Y_i = \beta_1 + \beta_2 D_i + u_i \quad (9.10.1)$$

where Y = hourly wage rate (\$) and $D = 1$ for female and 0 for male.

How do we interpret such a model? Assuming $E(u_i) = 0$, we obtain:

9.10 The Interpretation of Dummy Variables in Semi logarithmic Regressions

Wage function for male workers:

$$E(\ln Y_i | D_i = 0) = \beta_1 \quad (9.10.2)$$

Wage function for female workers:

$$E(\ln Y_i | D_i = 1) = \beta_1 + \beta_2 \quad (9.10.3)$$

Therefore, the intercept β_1 gives the *mean log hourly earnings* and the “slope” coefficient gives the difference in the mean log hourly earnings of male and females. This is a rather awkward way of stating things. But if we take the antilog of β_1 , what we obtain is *not* the mean hourly wages of male workers, but their **median** wages. As you know, *mean*, *median*, and *mode* are the three measures of central tendency of a random variable. And if we take the antilog of $(\beta_1 + \beta_2)$, we obtain the median hourly wages of female workers.

9.10 The Interpretation of Dummy Variables in Semi logarithmic Regressions

EXAMPLE 9.8 *Logarithm of Hourly Wages in Relation to Gender*

To illustrate Eq. (9.10.1), we use the data that underlie Example 9.2. The regression results based on 528 observations are as follows:

$$\begin{aligned}\widehat{\ln Y_i} &= 2.1763 - 0.2437D_i \\ t &= (72.2943)^* \quad (-5.5048)^* \\ R^2 &= 0.0544\end{aligned}\tag{9.10.4}$$

where * indicates p values are practically zero.

Taking the antilog of 2.1763, we find 8.8136 (\$), which is the median hourly earnings of male workers, and taking the antilog of $[(2.1763 - 0.2437) = 1.92857]$, we obtain 6.8796 (\$), which is the median hourly earnings of female workers. Thus, the female workers' median hourly earnings are lower by about 21.94 percent compared to their male counterparts $[(8.8136 - 6.8796)/8.8136]$.